Calculus - Week 1

Limits and continuity · Lectures 1–4

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The first question is not how to substitute a number into a formula, but what the values of the formula do near that number. A function can have a limit at a point even when its value there is missing or different. We start with that distinction, make it precise, and then use it to study continuity.

Preliminary

- $\mathbb{N}$ denotes the set of all positive integers.
- $\mathbb{Z}$ denotes the set of all integers.
- $\mathbb{Q}$ denotes the set of all rational numbers.
- $\mathbb{R}$ denotes the set of all real numbers.

Lecture 1: approaching a point

1. Key idea

We use limits to describe the tendency of a function as the variable approaches a given point, but not the value at the point.

2. Important examples

Example 1: Same limit, different value at the point

$$f(x) = \begin{cases} x^2 \sin \dfrac{1}{x}, & x \neq 0, \\ 0, & x = 0, \end{cases}$$

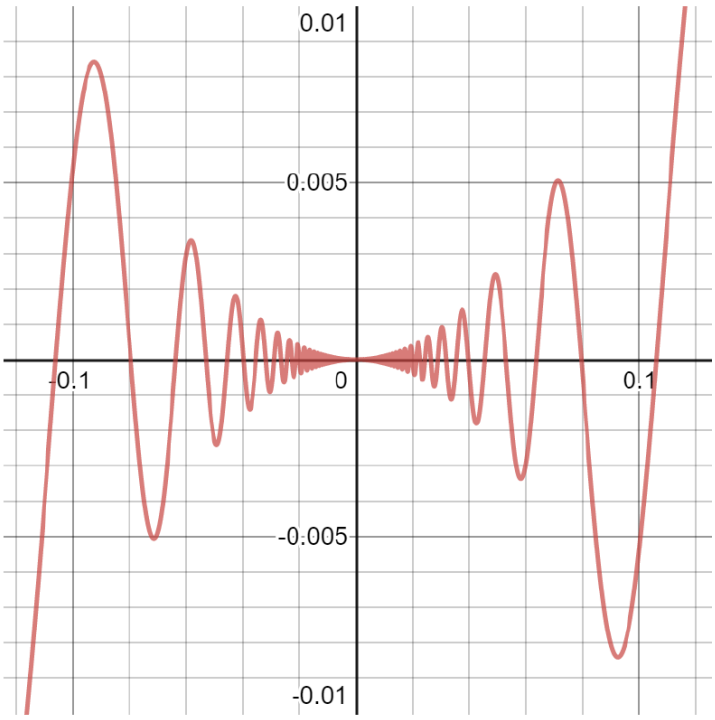


Figure: $f(x) = x^2 \sin \frac{1}{x}$ Zoom in

$$g(x) = \begin{cases} x \sin \dfrac{1}{x}, & x \neq 0, \\ 1, & x = 0. \end{cases}$$

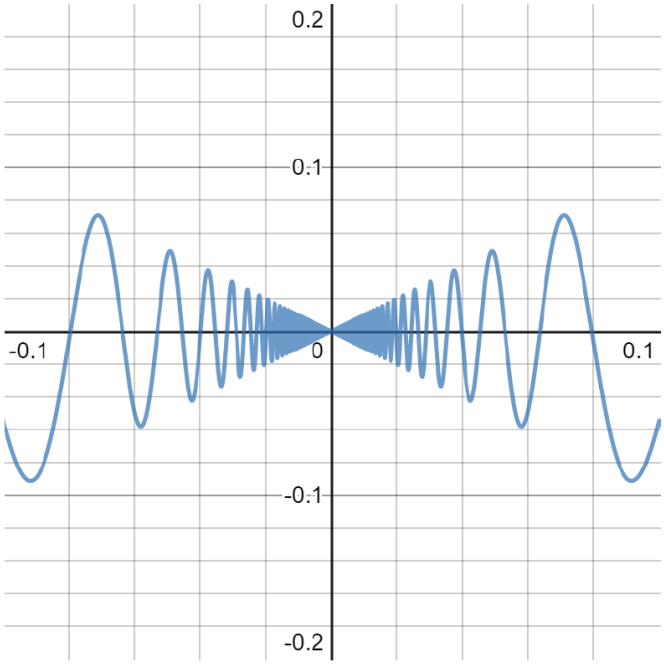


Figure: $g(x) = x \sin \frac{1}{x}$ Zoom in

For $x \neq 0$, the useful estimates are

$$|f(x)| \leq |x|^2, \qquad |g(x)| \leq |x|.$$

Therefore:

$$\lim_{x \to 0} f(x) = 0, \qquad \lim_{x \to 0} g(x) = 0.$$

My takeaway:

> If x gets close to 0, keeping $x \neq 0$, $f(x)$ approaches 0 with error $\leq |x|^2$; $g(x)$ approaches 0 with error $\leq |x|$.

Example 2: average velocity and instantaneous velocity

Average velocity:

$$\frac{\Delta s}{\Delta t} = \frac{f(t_2) - f(t_1)}{t_2 - t_1}.$$

With $h = t_2 - t_1$:

$$\frac{f(t_1 + h) - f(t_1)}{h}.$$

Instantaneous velocity at t_0:

$$v(t_0) = \lim_{h \to 0} \frac{f(t_0 + h) - f(t_0)}{h} = \lim_{t \to t_0} \frac{f(t) - f(t_0)}{t - t_0}.$$

Informal definition of limit

> Assume $f(x)$ is defined on an open interval about c, except possibly at c itself. If $f(x)$ is arbitrarily close to L for all x sufficiently close to c other than c itself, then

$$\lim_{x \to c} f(x) = L.$$

This is read "the limit of $f(x)$ as x approaches c is L."

3. Limit calculation techniques

Algebraic simplification

$$\lim_{x \to 1} \frac{x^2 - 1}{x - 1} = \lim_{x \to 1} (x + 1) = 2.$$

Rationalizing the denominator

$$\lim_{x \to 1} \frac{\sqrt{x} - 1}{x - 1} = \lim_{x \to 1} \frac{1}{\sqrt{x} + 1} = \frac{1}{2}.$$

Remark: Simplification when calculating limits.

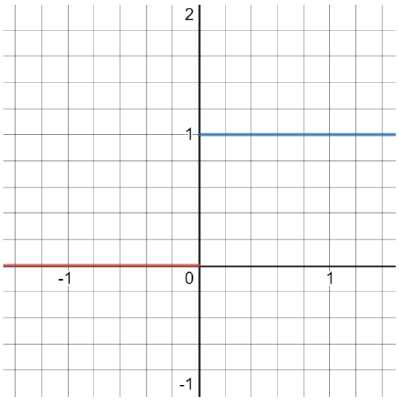
General strategy:

1. Check direct substitution.
2. Factor or rationalize if necessary.
3. Simplify, then evaluate the limit.

Class 2: Describing the behavior of a function as $x \to 0$

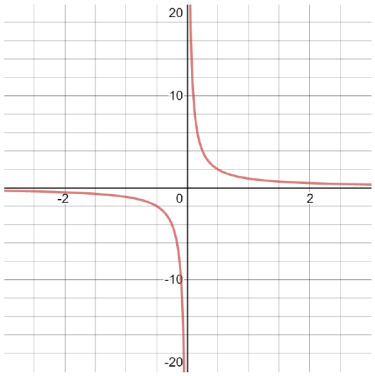
1. Jump

Unit step function $U(x) = \begin{cases} 0, & x < 0, \\ 1, & x \geq 0. \end{cases}$



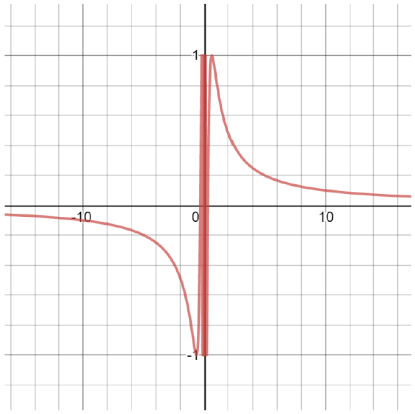
2. Unbounded

Unbounded function $g(x) = \begin{cases} \frac{1}{x}, & x \neq 0, \\ 0, & x = 0, \end{cases}$



3. Oscillation

Oscillating function $h(x) = \begin{cases} \sin\frac{1}{x}, & x \neq 0 \\ 0 & x = 0 \end{cases}$



4. When a limit does not exist

Type	Behavior near the point	Example
Jump	Different left and right limits	Unit step function
Unbounded	Values grow without bound	$1/x$ near zero
Oscillation	No single limiting value	$\sin(1/x)$ near zero

Lecture 2: the definition and how to use it

Definition of limit (Informal)

$$g(x) = \begin{cases} 2x, & \text{if } x \neq 4, \\ 0, & \text{if } x = 4. \end{cases}$$

By the informal description of a limit: if x gets sufficiently close to 4, but is not equal to 4, then $g(x)$ gets arbitrarily close to 8.

We need to show that the gap between $g(x)$ and $L = 8$, namely $|g(x) - 8|$, can be made as small as we choose if x is kept close enough to $c = 4$, but is not equal to 4.

Definition

Let $f(x)$ be defined on an open interval I about c, except possibly at c itself. We say that the limit of $f(x)$ as x approaches c is the number L, and write

$$\lim_{x \to c} f(x) = L,$$

if, for every $\varepsilon > 0$, there exists a corresponding $\delta > 0$ such that

$$|f(x) - L| < \varepsilon \quad \text{whenever } x \in I \text{ and } 0 < |x - c| < \delta.$$

Simpler writing

We say $\lim_{x \to c} f(x) = L$ if, for any $\varepsilon > 0$, there exists $\delta > 0$ such that

$$\text{if } x \in I, \quad 0 < |x - c| < \delta, \quad \text{then } |f(x) - L| < \varepsilon.$$

Personal interpretation

- ε: how close we require $f(x)$ to be to L.
- δ: how close x needs to be to c in order to satisfy that requirement.
- $|f(x) - L| < \varepsilon$: $f(x)$ lies within an ε-neighborhood of L.
- $0 < |x - c| < \delta$: x lies within a δ-neighborhood of c, but $x \neq c$.

The key idea is

$$\boxed{\text{Given any } \varepsilon > 0, \text{ we can find a } \delta > 0}$$

such that

$$x \text{ is sufficiently close to } c \quad \Longrightarrow \quad f(x) \text{ is within } \varepsilon \text{ of } L.$$

In other words, ε is the required accuracy for the output, while δ is how much accuracy we need for the input.

Thus, the ε-δ definition gives a precise mathematical meaning to the intuitive statement: if x is sufficiently close to c, then $f(x)$ is sufficiently close to L.

Useful remarks

The order of the quantifiers matters. The number δ may depend on ε, on the function, and on the point c, but not on the particular x that will later be chosen. We need one radius that works for every point in the punctured interval. Making δ smaller never spoils a valid choice.

The condition $0 < |x - c|$ deliberately excludes c. Continuity will put the value at c back into the discussion.

We shall often use the following arguments:

1. Let a be a real number. If, for every $\varepsilon > 0$, we have $a \leq \varepsilon$, then $a \leq 0$.

2. Let a be a real number. If, for every $\varepsilon > 0$, we have

$$-\varepsilon \leq a \leq \varepsilon,$$

then $a = 0$.

Example: revisiting $g(x)$

Consider

$$g(x) = \begin{cases} 2x, & \text{if } x \neq 4, \\ 0, & \text{if } x = 4. \end{cases}$$

Show that

$$\lim_{x \to 4} g(x) = 8.$$

Analysis. For every $\varepsilon > 0$, we need

$$|g(x) - 8| = |2x - 8| < \varepsilon, \qquad x \neq 4,$$

which requires

$$0 < |x - 4| < \frac{\varepsilon}{2}.$$

Proof. Given arbitrary $\varepsilon > 0$, take

$$\delta = \frac{\varepsilon}{2}.$$

Then

$$|g(x) - 8| = |2x - 8| = 2|x - 4| < 2\delta = \varepsilon$$

whenever $0 < |x - 4| < \delta$. By the definition of limit,

$$\lim_{x \to 4} g(x) = 8.$$

Exercise

Prove that

$$\lim_{x\to 1}\frac{1}{x}=1.$$

Hint. Restrict x by taking $x>\frac{1}{2}$. Then

$$\left|\frac{1}{x}-1\right|=\frac{|x-1|}{|x|}<2|x-1|.$$

Solution. Given $\varepsilon>0$, choose $\delta=\min\{1/2,\varepsilon/2\}$. If $0<|x-1|<\delta$, then $x>1/2$ and the displayed estimate is less than ε. The first restriction keeps the denominator away from zero; the second controls the error.

Exercise 1.1.5 revisited

Using the definition of limit, prove that the following functions have no limit as $x\to 0$.

(a)

$$U(x)=\begin{cases}0, & x<0,\\ 1, & x\geq 0.\end{cases}$$

(b)

$$g(x)=\begin{cases}\dfrac{1}{x}, & x\neq 0,\\ 0, & x=0.\end{cases}$$

(c)

$$f(x)=\begin{cases}0, & x\leq 0,\\ \sin\dfrac{1}{x}, & x>0.\end{cases}$$

Tip. Suppose the limit exists and derive a contradiction.

For (b)

Suppose

$$\lim_{x\to 0}\frac{1}{x}=L.$$

Use the definition with $\varepsilon_0 = 1$. It would give a radius $\delta > 0$ on which $|1/x - L| < 1$. Choose

$$0 < x < \min\{\delta, 1/(|L| + 2)\}.$$

Then $1/x > |L| + 2$, so $|1/x - L| \geq 1/x - |L| > 2$, a contradiction.

Solutions to (a) and (c). For the step function, points on the two sides have values 0 and 1. If a limit L existed, choose $\varepsilon = 1/3$ and a common radius. These two values would satisfy $1 \leq |1 - L| + |L| < 2/3$, which is impossible.

For (c), positive points $x_n = (\pi/2 + 2n\pi)^{-1}$ and $y_n = (3\pi/2 + 2n\pi)^{-1}$ approach zero and give values 1 and -1. A proposed limit would have to be within $1/2$ of both values for all sufficiently large n, again impossible. These sequences are a convenient way to choose points in every sufficiently small neighborhood; a formal sequence criterion comes in Week 2.

Lecture 3: limit laws and one-sided limits

Date: 2026.9.10

Theorem 1.1.3 (Limit Laws)

Assume that

$$\lim_{x \to c} f(x) = L \qquad \text{and} \qquad \lim_{x \to c} g(x) = M.$$

(a) Sum Rule

$$\lim_{x \to c} \big(f(x) + g(x)\big) = \lim_{x \to c} f(x) + \lim_{x \to c} g(x).$$

Proof. Let $\varepsilon > 0$. Choose $\delta_1, \delta_2 > 0$ so that $|f(x) - L| < \varepsilon/2$ and $|g(x) - M| < \varepsilon/2$. For $\delta = \min\{\delta_1, \delta_2\}$,

$$|(f(x) + g(x)) - (L + M)| \leq |f(x) - L| + |g(x) - M| < \varepsilon.$$

(b) Product Rule

$$\lim_{x \to c} \big(f(x) \cdot g(x)\big) = \lim_{x \to c} f(x) \cdot \lim_{x \to c} g(x).$$

Proof. Since $f(x) \to L$, it is bounded near c; take a neighborhood where $|f(x)| \leq |L| + 1$. Then

$$|f(x)g(x) - LM| \leq |f(x)||g(x) - M| + |M||f(x) - L|,$$

For an explicit choice, first restrict to $|f(x) - L| < 1$. Also impose

$$|g(x) - M| < \frac{\varepsilon}{2(|L| + 1)}, \qquad |f(x) - L| < \frac{\varepsilon}{2(|M| + 1)}.$$

Take the minimum of the three radii. The two terms in the estimate are then respectively less than $\varepsilon/2$ and at most $\varepsilon/2$. This also covers $L = 0$ or $M = 0$, without division by either number.

(c) Quotient Rule

If $M \neq 0$, then

$$\lim_{x \to c} \frac{f(x)}{g(x)} = \frac{\lim_{x \to c} f(x)}{\lim_{x \to c} g(x)}.$$

Proof. Since $g(x) \to M \neq 0$, eventually $|g(x)| > |M|/2$. Hence

$$\left| \frac{1}{g(x)} - \frac{1}{M} \right| = \frac{|g(x) - M|}{|g(x)||M|} \leq \frac{2|g(x) - M|}{|M|^2} \to 0.$$

Thus $1/g(x) \to 1/M$, and the quotient rule follows from the product rule.

Remark: Does the Converse Hold?

The Sum Rule requires that both limits on the right-hand side exist. In general, the converse is false: the existence of

$$\lim_{x \to c} (f(x) + g(x))$$

does not imply that both $\lim_{x \to c} f(x)$ and $\lim_{x \to c} g(x)$ exist.

Counterexample

Let

$$f(x) = \sin \frac{1}{x}, \qquad g(x) = -\sin \frac{1}{x}.$$

Consider $x \to 0$. Although

$$\lim_{x \to 0} \sin \frac{1}{x}$$

does not exist, we have

$$f(x) + g(x) = 0.$$

Therefore,

$$\lim_{x\to 0}(f(x)+g(x))=0$$

exists. Thus,

$$\boxed{\lim(f+g)\text{ exists} \not\Longrightarrow \lim f, \lim g \text{ both exist}}$$

The problematic behavior of two functions can cancel each other out.

Exercise 1.1.9 (Limits of Polynomials)

If

$$P(x)=\sum_{k=0}^{n} a_k x^k = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0,$$

then

$$\lim_{x\to c} P(x) = \sum_{k=0}^{n} a_k c^k = P(c).$$

Exercise 1.1.10 (Limits of Rational Functions)

If $P(x)$ and $Q(x)$ are polynomials and $Q(c)\neq 0$, then

$$\lim_{x\to c}\frac{P(x)}{Q(x)}=\frac{P(c)}{Q(c)}.$$

Theorem 1.1.4 (The Sandwich Theorem)

Suppose that

$$g(x)\leq f(x)\leq h(x)$$

for all x in some open interval containing c, except possibly at $x=c$. Suppose also that

$$\lim_{x\to c} g(x) = \lim_{x\to c} h(x) = L.$$

Then

$$\lim_{x\to c} f(x) = L.$$

Proof. Given $\varepsilon > 0$, choose $\delta > 0$ so that $L - \varepsilon < g(x)$ and $h(x) < L + \varepsilon$ whenever $0 < |x - c| < \delta$. The inequalities $g(x) \leq f(x) \leq h(x)$ then imply $|f(x) - L| < \varepsilon$.

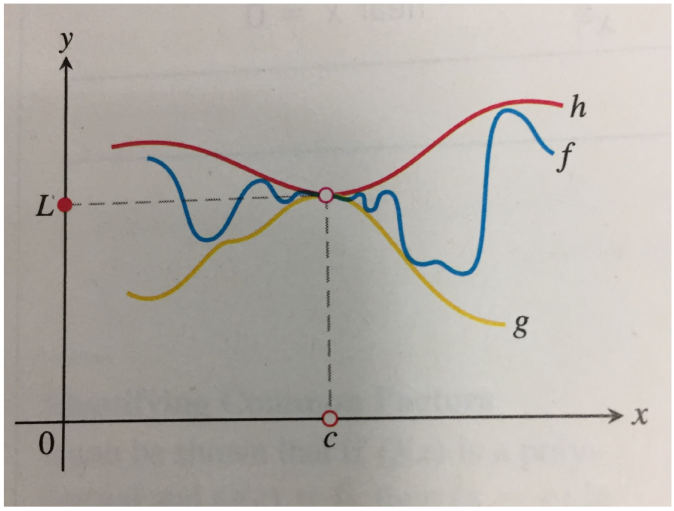


Figure: The Sandwich Theorem: $g(x) \leq f(x) \leq h(x)$

Exercise 1.1.12

Assume I is an open interval, $c \in I$, and

$$f(x) \leq b \qquad \text{for all } x \in I \setminus \{c\}.$$

Then

$$\lim_{x \to c} f(x) \leq b,$$

provided the limit exists.

Proof. Write $L = \lim_{x \to c} f(x)$. If $L > b$, take $\varepsilon = (L - b)/2$. Near c the definition gives $f(x) > L - \varepsilon = (L + b)/2 > b$, contradicting the assumed upper bound.

Exercise 1.1.13 (Order-preserving Property of Limits)

If

$$f(x) \leq g(x)$$

for all x in some open interval containing c, except possibly at $x = c$, and both limits exist, then

$$\lim_{x \to c} f(x) \leq \lim_{x \to c} g(x).$$

Limits preserve the non-strict inequality $\leq$, but may not preserve the strict inequality $<$. For example, let

$$f(x) = x^2, \qquad g(x) = 2x^2.$$

For $x \neq 0$, $f(x) < g(x)$, but

$$\lim_{x\to 0} x^2 = \lim_{x\to 0} 2x^2 = 0.$$

Lemma 1.1.5

For θ measured in radians,

$$-|\theta| \leq \sin\theta \leq |\theta|, \qquad -|\theta| \leq 1 - \cos\theta \leq |\theta|.$$

In particular, for $0 < \theta \leq \dfrac{\pi}{2}$,

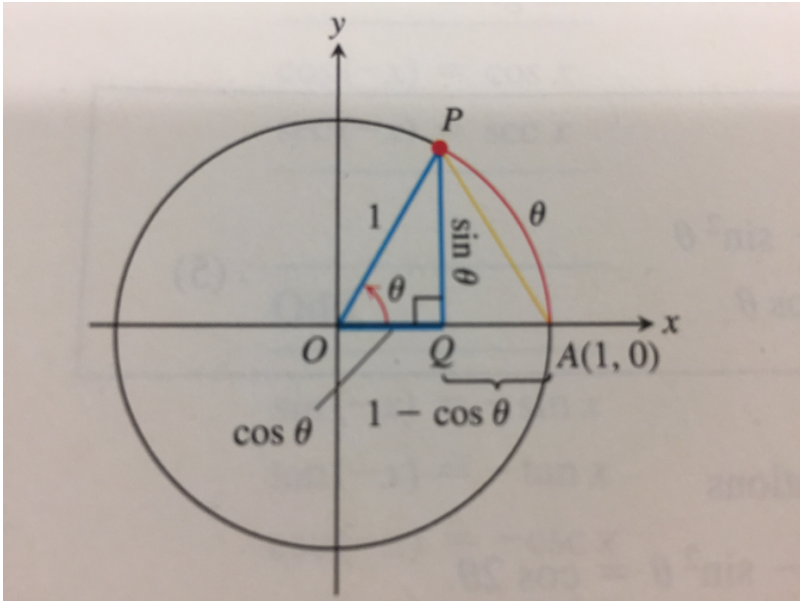
$$0 < \sin\theta < \theta.$$

Geometric proof. For $0 < \theta < \pi/2$, take $A = (1, 0)$ and $P = (\cos\theta, \sin\theta)$ on the unit circle. The chord AP is shorter than the arc of length θ. Therefore

$$\sin^2\theta + (1 - \cos\theta)^2 = |AP|^2 < \theta^2.$$

Each nonnegative term gives $|\sin\theta| < \theta$ and $1 - \cos\theta < \theta$ in this range. At $\theta = \pi/2$ the claimed inequalities also hold. Symmetry handles negative angles. For $|\theta| \geq \pi/2$, use $|\sin\theta| \leq 1 \leq |\theta|$ and $1 - \cos\theta = 2\sin^2(\theta/2) \leq 2|\sin(\theta/2)| \leq |\theta|$; the last sine bound is already established globally.

This argument precedes differentiation. Using the mean value theorem here would reverse the logical order: the derivatives of sine and cosine will themselves use the trigonometric limit below.



One-sided Limits

Left-hand limit

$$\lim_{x \to c^-} f(x) = L$$

if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$c - \delta < x < c \quad \Longrightarrow \quad |f(x) - L| < \varepsilon.$$

Right-hand limit

$$\lim_{x \to c^+} f(x) = L$$

if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$c < x < c + \delta \quad \Longrightarrow \quad |f(x) - L| < \varepsilon.$$

The two-sided limit exists if and only if the one-sided limits both exist and are equal:

$$\lim_{x \to c} f(x) = L \quad \Longleftrightarrow \quad \lim_{x \to c^-} f(x) = \lim_{x \to c^+} f(x) = L.$$

Examples at $x \to 0$

Example 1: The 0-1 step function.

$$U(x) = \begin{cases} 0, & x < 0, \\ 1, & x \geq 0. \end{cases}$$

Then

$$\lim_{x \to 0^-} U(x) = 0, \qquad \lim_{x \to 0^+} U(x) = 1,$$

so the two-sided limit does not exist.

Example 2: The function $1/x$.

$$\lim_{x \to 0^-} \frac{1}{x} = -\infty, \qquad \lim_{x \to 0^+} \frac{1}{x} = +\infty.$$

Thus, $1/x$ has no finite two-sided limit at 0.

Example 3: The oscillating function $\sin(1/x)$**.**

$$\lim_{x\to 0^-} \sin\frac{1}{x} \quad \text{and} \quad \lim_{x\to 0^+} \sin\frac{1}{x}$$

both do not exist.

The density of $\mathbb{Q}$ **in** $\mathbb{R}$**.** We say that $\mathbb{Q}$ is dense in $\mathbb{R}$ if every non-empty open interval contains a rational number. For any $a, b \in \mathbb{R}$ with $a < b$, there exists $q \in \mathbb{Q}$ such that

$$a < q < b.$$

Equivalently, for every $x \in \mathbb{R}$ and every $\varepsilon > 0$, there exists $q \in \mathbb{Q}$ such that

$$|q - x| < \varepsilon.$$

The irrational numbers are also dense in $\mathbb{R}$.

Example 4: The Dirichlet function.

$$D(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \in \mathbb{R} \setminus \mathbb{Q}. \end{cases}$$

Every interval to the left or right of 0 contains both rational and irrational numbers. Hence,

$$\lim_{x\to 0^-} D(x) \quad \text{and} \quad \lim_{x\to 0^+} D(x)$$

both do not exist.

Theorem 1.1.6

Suppose that $f(x)$ is defined on an open interval containing c, except possibly at c itself. Then $f(x)$ has a limit as x approaches c if and only if its left-hand and right-hand limits both exist and are equal:

$$\lim_{x\to c} f(x) = L \quad \Longleftrightarrow \quad \lim_{x\to c^-} f(x) = \lim_{x\to c^+} f(x) = L.$$

Proof

If $\lim_{x\to c} f(x) = L$, the ε-δ condition immediately implies the same estimate for $x < c$ and for $x > c$. Therefore both one-sided limits equal L.

Conversely, suppose both one-sided limits equal L. Given $\varepsilon > 0$, choose $\delta_1, \delta_2 > 0$ from the left- and right-hand definitions and let

$$\delta = \min\{\delta_1, \delta_2\}.$$

If $0 < |x - c| < \delta$, then either $c - \delta < x < c$ or $c < x < c + \delta$. In either case,

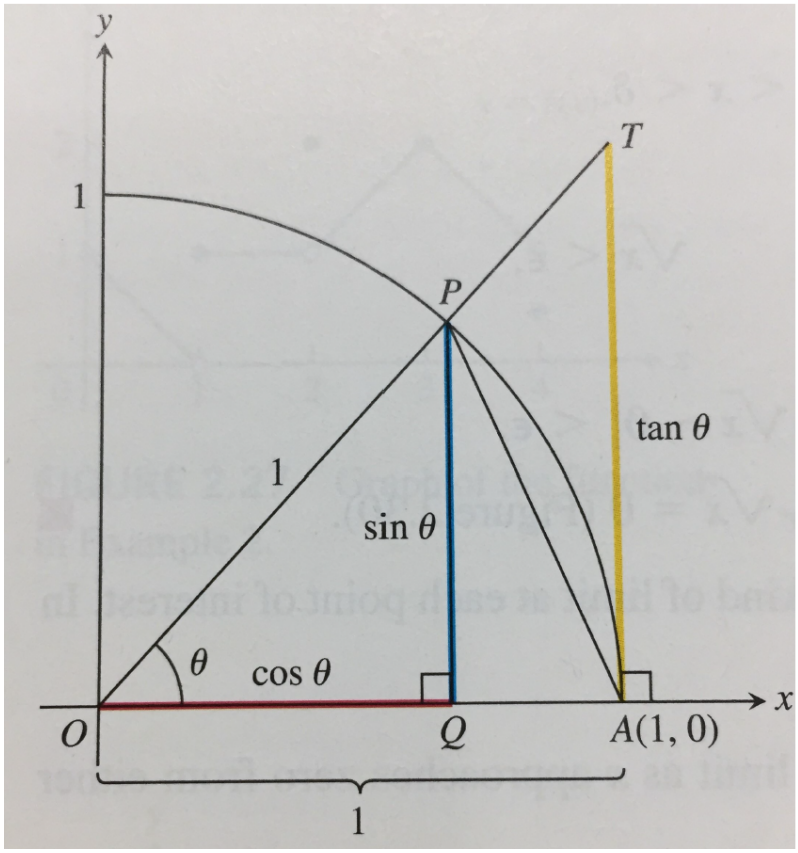
$$|f(x) - L| < \varepsilon.$$

Thus $\lim_{x \to c} f(x) = L$.

Theorem 1.1.7

For θ measured in radians,

$$\lim_{\theta \to 0} \frac{\sin \theta}{\theta} = 1.$$



Proof

For $0 < \theta < \frac{\pi}{2}$,

$$\cos \theta \leq \frac{\sin \theta}{\theta} \leq 1.$$

Since $\lim_{\theta \to 0} \cos \theta = 1$, the Sandwich Theorem gives

$$\lim_{\theta \to 0^+} \frac{\sin \theta}{\theta} = 1.$$

For $\theta < 0$,

$$\frac{\sin \theta}{\theta} = \frac{\sin(-\theta)}{-\theta},$$

so the left-hand limit is also 1. Therefore,

$$\boxed{\lim_{\theta \to 0} \frac{\sin \theta}{\theta} = 1}.$$

Consequences

For any constant $k \neq 0$,

$$\lim_{\theta \to 0} \frac{\sin(k\theta)}{\theta} = k \lim_{\theta \to 0} \frac{\sin(k\theta)}{k\theta} = k.$$

Also,

$$\lim_{x \to 0} \frac{\sin(\sin x)}{\sin x} = 1,$$

and, for n nested sine functions,

$$\lim_{x \to 0} \frac{\sin\big(\sin(\cdots \sin x \cdots)\big)}{x} = 1.$$

Proof of $\lim_{x \to 0} \dfrac{\sin(\sin x)}{x} = 1$

Since $\lim_{x \to 0} \sin x = 0$, Theorem 1.1.7 gives

$$\lim_{x \to 0} \frac{\sin(\sin x)}{\sin x} = 1.$$

Moreover,

$$\frac{\sin(\sin x)}{x} = \frac{\sin(\sin x)}{\sin x} \cdot \frac{\sin x}{x}.$$

Therefore,

$$\begin{aligned} \lim_{x \to 0} \frac{\sin(\sin x)}{x} &= \left(\lim_{x \to 0} \frac{\sin(\sin x)}{\sin x} \right) \left(\lim_{x \to 0} \frac{\sin x}{x} \right) \\ &= 1 \cdot 1 = 1. \end{aligned}$$

Lecture 4: continuity

Example 1.1.19

$$f(x) = \begin{cases} x, & x < 0, \\ 1, & 0 \le x < 1, \\ 2, & x = 1, \\ x, & x > 1. \end{cases}$$

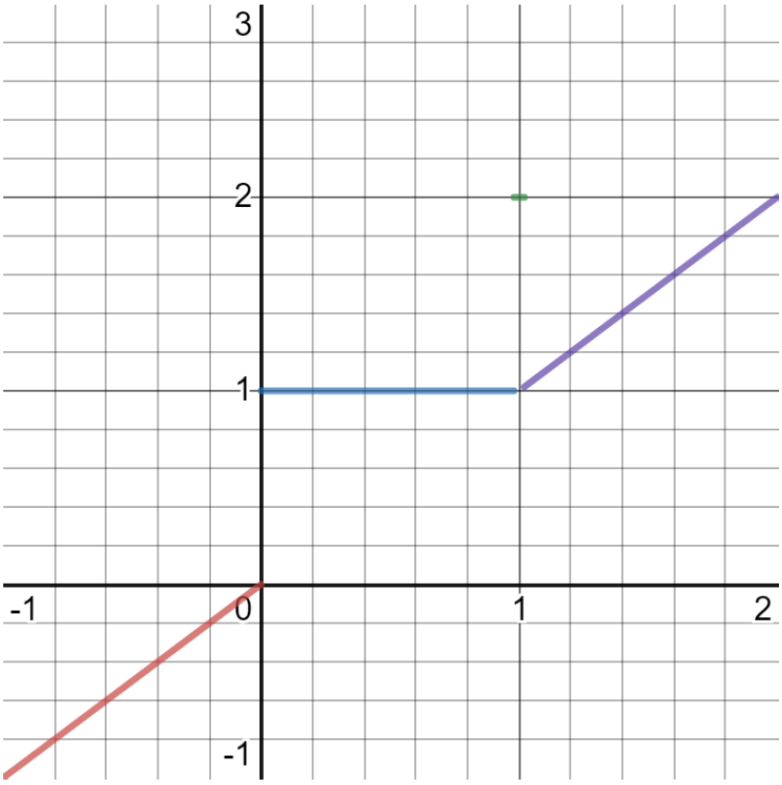


Figure: jump discontinuity

Definition 1.1.8 (Continuity at a Point)

Let c be a real number that is either an interior point or an endpoint of an interval in the domain of f.

(a) Interior point

f is continuous at c if

$$\lim_{x \to c} f(x) = f(c).$$

(b) Right-continuity

f is right-continuous at c (or continuous from the right) if

$$\lim_{x \to c^+} f(x) = f(c).$$

(c) Left-continuity

f is left-continuous at c (or continuous from the left) if

$$\lim_{x \to c^-} f(x) = f(c).$$

Additional examples from the lectures

Uniqueness of the limit

Proposition. A function cannot have two different limits at the same point. If $\lim_{x \to c} f(x) = L$ and $\lim_{x \to c} f(x) = K$, then $L = K$.

Proof. Suppose $L \neq K$ and choose $\varepsilon = |L - K|/3$. The two ε-neighborhoods of L and K are disjoint, but the definition of limit would force $f(x)$ to lie in both for x sufficiently close to c. This contradiction proves $L = K$.

A useful ε-δ example

To prove $\lim_{x \to 2} x^2 = 4$, observe

$$|x^2 - 4| = |x - 2||x + 2|.$$

If $|x - 2| < 1$, then $|x + 2| < 5$. Thus it is enough to take $\delta = \min\{1, \varepsilon/5\}$.

Power and root rules

The limit laws imply

$$\lim_{x \to c} [f(x)]^n = \left(\lim_{x \to c} f(x) \right)^n.$$

If $f(x) \geq 0$ near c and the right-hand limit is nonnegative, then

$$\lim_{x \to c} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \to c} f(x)}.$$

For example,

$$\lim_{x\to 4}\sqrt{x}=2, \qquad \lim_{x\to 1}\frac{x^3+2x}{x+1}=\frac{3}{2}.$$

One-sided limit example

For $f(x)=|x|/x$,

$$\lim_{x\to 0^-} f(x)=-1, \qquad \lim_{x\to 0^+} f(x)=1,$$

so the two-sided limit does not exist. This is the simplest model of a jump discontinuity.

Continuity test and types of discontinuity

At an interior point c, continuity requires all three conditions:

1. $f(c)$ is defined;
2. $\lim_{x\to c} f(x)$ exists;
3. $\lim_{x\to c} f(x)=f(c)$.

For the piecewise function in Example 1.1.19, the point 0 is a jump discontinuity because the one-sided limits are 0 and 1. At 1, the two-sided limit is 1 but $f(1)=2$, so the discontinuity is removable: changing the value at 1 to 1 would make the function continuous there.

Riemann-type example

On $[0,1]$, put $R(p/q)=1/q$ when p/q is in lowest terms with $q>0$, and put $R(x)=0$ for irrational x. In particular $R(0)=R(1)=1$. This is also called Thomae's function. The lecture extends it periodically outside $[0,1]$.

At a rational point. If $c=p/q$, irrational points occur arbitrarily close to c and have value zero, whereas $R(c)=1/q$. Taking $\varepsilon=1/(2q)$ disproves continuity. At 0 or 1, use points from inside the interval.

At an irrational point. Fix an irrational $c\in(0,1)$ and $\varepsilon>0$. Choose an integer $N>1/\varepsilon$. There are only finitely many reduced fractions in $[0,1]$ with denominator at most N; call this finite set E_N. None equals c, so the distance from c to E_N is positive. Choose δ smaller than that distance and than c and $1-c$.

If $|x-c|<\delta$, either x is irrational and $R(x)=0$, or its reduced denominator is greater than N and $R(x)<1/N<\varepsilon$. Since $R(c)=0$, this is exactly continuity at c.

The useful idea is that the tall spikes form a finite set once a height is fixed. Density of the rationals by itself does not rule out continuity.

Continuity on an interval and continuous extension

Continuity on $[a, b]$ means ordinary continuity on (a, b), right-continuity at a, and left-continuity at b. Use the corresponding endpoint convention for a half-open interval.

If f is continuous on an interval except at c and $\lim_{x\to c} f(x) = L$ exists, defining $f(c) = L$ gives its unique continuous extension at c. For example, the function $\sin x/x$ extends to a continuous function at zero by assigning the value 1. An infinite or oscillating limit cannot be repaired by changing just one function value.

Review problems

1. Prove $\lim_{x\to 2} x^2 = 4$ without quoting the polynomial limit law. Explain why the restriction $|x - 2| < 1$ is useful.
2. Give functions whose sum has a limit but neither summand does. Then give an example showing that strict inequalities need not survive a limit.
3. Compare $\sin(1/x)$, $x\sin(1/x)$ and $x^2\sin(1/x)$ near zero. Which can be extended continuously? Differentiability will be a separate question in Week 3.

Lecture sources

Xing-Bin Pan, MAT1011 lecture slides, Lectures 1–4: limits and limit language; limit laws and the geometric trigonometric limit; continuity and the Riemann function (Lecture 4, pp.20–29). The original diagrams and lecture numbering are retained. Course assessment recorded in the original draft: class 5%, homework 15%, quizzes 20%, midterm 20%, final 40%.